# Data-Backed Journal-Style Replication of *Overcoming Local Search Through Alliances and Mobility*

Replication Paper Project

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**Abstract**

This paper presents a methodology-specific local analogue of Overcoming Local Search Through Alliances and Mobility using only processed datasets and result files available in the workspace. The replication preserves the source paper's core empirical logic while making every local substitution explicit. The paper is therefore reported as methodology-specific local analogue rather than as a source-data replay. The current local batch contains 23,901 observations across 184 entities, and the batch headline estimate is 0.0722 for Mobility Bridge x Technological Distance with p = 0.002. The executable analogue emphasizes whether bridge mechanisms become more valuable as recombination distance increases, so the alliance-by-distance and mobility-by-distance interactions are central. The contribution is a journal-style replication manuscript that separates direct source claims from executable local evidence.

**Keywords:** local search; alliances; mobility; technological distance; patent knowledge flows; local calculated analogue

## 1 Introduction

This paper begins from a paper-first reading of Overcoming Local Search Through Alliances and Mobility and treats the source article as an empirical argument that has to be rebuilt before it can be judged. The present manuscript is classified as methodology-specific local analogue. A paper-first restatement of the empirical question is as follows: The source paper asks how firms overcome

local search and reach distant knowledge through alliances and mobility (**????**). The replication matters because the paper is a core statement of how firms escape localized search without abandoning their existing knowledge base entirely. The original article does not rest on a descriptive regularity alone. It ties a theory about search, differentiation, timing, or organizational learning to a specific unit of analysis, sample frame, and estimand. A careful replication therefore has to recover the structure of that logic before any local coefficients are interpreted (**?????**). The executable evidence base used in this manuscript contains 23,901 observations across 184 entities and spans 1986 to 2016 in the local executable panel. Those numbers are not filler. They define the actual inferential scope of the local reconstruction and set the boundary on what can be claimed honestly from the available datasets (**??????**). The paper's contribution is a transparent bridge-mechanism analogue that can be estimated from the current processed patent and citation-network stack. The paper is written to read like a formal journal article rather than an internal workflow memo. The main text develops the theoretical logic, data reconstruction, derivations, empirical design, and results in continuous prose, while the appendices preserve the audit trail required for reruns and later extensions (**??????**). The paper sits in a top-journal literature on localized search, interfirm knowledge flows, and the mechanisms that help firms bridge distance. (**?????????**). That literature matters because the local rerun is about mechanism transportability: whether the value of bridge mechanisms survives after the tie architecture is translated into processed network and recombination variables. (**????????????**). The present manuscript therefore treats the source paper as one contribution inside a broader top-journal stream on exploration, recombination, performance measurement, alliance-based learning, inventive search, technological discontinuity, and patent-based inference. The literature review is intentionally broad because each replication paper should stand on a serious journal-level reference base rather than on the source article alone. (**?????????**). The paper-specific review therefore deepens the network and knowledge-flow conversation rather than relying only on generic exploration references. (**???**). A second topic-specific literature block is added to sharpen the manuscript's empirical boundary and identification problem. (**??**). A final topic-specific literature block keeps the bibliography tied to the paper's actual mechanism rather than to a generic replication template. (**???**).

# 2 Original Study, Theory, and Replication Boundary

The focal source article is (**?**). The original study can be summarized as a mechanism paper with a sharp empirical core. The published paper studies alliances and inventor mobility as mechanisms that help firms overcome local search. Technological distance is a central moderator in the argument. The empirical outcome concerns knowledge flows and innovative usefulness rather than simple patent volume. That combination matters because the source article asks readers to believe not merely that two variables move together, but that a particular organizational or strategic mechanism explains why the relationship should appear in the data (**????????**). In a setting like this, faithful replication requires more than reusing a title and rerunning a regression. The theoretical contribution lives in the way constructs are defined, how they enter the estimand, and which comparison the model is supposed to adjudicate. For that reason, this manuscript first reconstructs the published theory and only then turns to the local executable analogue (**????**). The replication boundary is therefore explicit. Direct source facts are carried over when they can be read from the published article itself. Local replication decisions appear only when the current workspace must translate the source design into verified processed datasets. The goal is disciplined mechanism recovery, not rhetorical imitation that quietly changes the design (**????????**).

## 2.1 Original Paper Methodology

A faithful evaluation of Rosenkopf and Almeida (2003) starts with the published empirical design rather than with the local rerun. The source article first establishes geographic and technological localization of knowledge in the semiconductor industry and then evaluates whether alliances and the mobility of active inventors bridge otherwise distant knowledge contexts (**????**). At the level of sample construction, the source article works from The published sample follows a cohort of semiconductor firms and studies interfirm knowledge flows in that industry setting. At the level of measurement, it relies on The published design uses direct alliance and inventor-mobility evidence together with semiconductor patent citation patterns to trace interfirm knowledge flows. These details matter because they determine what the published paper actually observes rather than what a later analogue can observe (**????**). At the level of estimation, the published paper uses patent-citation-based models of interfirm knowledge flows that relate those flows to alliances,

inventor mobility, and technological distance in semiconductors. The critical published comparison is whether alliances and inventor mobility become more useful as technological distance rises, conditional on the original semiconductor patent-citation setting. Reading the source methodology this way prevents the current manuscript from quietly replacing the original design with one generic fixed-effects story (**????**).

| DesignElement | SourcePaperDesign |
|---|---|
| Published empirical setting | The published sample follows a cohort of semiconductor firms and studies interfirm knowledge flows in that industry setting. |
| Published measurement architecture | The published design uses direct alliance and inventor-mobility evidence together with semiconductor patent citation patterns to trace interfirm knowledge flows. |
| Published estimator | patent-citation-based models of interfirm knowledge flows that relate those flows to alliances, inventor mobility, and technological distance in semiconductors |
| Published identification and comparison rule | The critical published comparison is whether alliances and inventor mobility become more useful as technological distance rises, conditional on the original semiconductor patent-citation setting. |

Published empirical design reconstructed from the source paper before local estimation begins.

## 2.2 Replication Status and Boundary

Replication status for this paper: Methodology-specific local analogue (**????**). This paper is a methodology-specific local analogue because the original article directly observes alliances and inventor mobility in semiconductors, whereas the local rerun maps those mechanisms onto citation-network bridges and outside-the-firm search cases. The current data can test the bridge logic, but not replay the original mechanism archive one-for-one (**????**).

## 2.3 Direct Source Facts Versus Replication Decisions

The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Alliances and mobility are the two bridge mechanisms. The local reconstruction therefore adopts the following bounded decision: Alliance bridges are proxied with rolling citation-network ties and mobility bridges with outside-the-firm search cases. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (**????**). The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Technological distance moderates the value of those bridges. The local reconstruction therefore adopts the following bounded decision: Technological distance is mapped to recombination breadth in the local patent panel. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (**????**). The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that The outcome is about knowledge-usefulness rather than raw output. The local reconstruction therefore adopts the following bounded decision: Patent-level inventive impact is used as the executable outcome. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit

local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (**????**).

| DirectSourceFact | LocalReplicationDecision |
|---|---|
| Alliances and mobility are the two bridge mechanisms. | Alliance bridges are proxied with rolling citation-network ties and mobility bridges with outside-the-firm search cases. |
| Technological distance moderates the value of those bridges. | Technological distance is mapped to recombination breadth in the local patent panel. |
| The outcome is about knowledge-usefulness rather than raw output. | Patent-level inventive impact is used as the executable outcome. |

Direct source facts and explicit local execution decisions carried in the body of the paper.

## 3 Data, Sample, and Measures

The local evidence chain is deliberately modular. The source paper is treated as the authority for the research question, theoretical mechanism, and published design. Estimation, by contrast, proceeds only through the verified local files. The required dataset types in this paper are Patent-level recombination panel with organizational and field experience counts from the local USPTO focal-patent workflow. Rolling firm-year interfirm citation-link network measures built from processed backward patent ties. Patent-level inventive impact and patenting controls from the cleaned reproduction dataset (**???**). Operationally, the reconstruction relies on four evidence layers: the analysis panel, the coefficient export, the sample summary, and, when available, the fit summary. The resulting executable panel supplies 23,901 observations across 184 entities and covers 1986 to 2016 in the local executable panel. Every numeric claim in the results section is intended to trace back to one of those verified evidence layers rather than to a hand-copied number from the article

(**??????**). This data-lineage discipline matters for two reasons. First, it prevents the manuscript from smuggling unverified preprocessing steps into the design. Second, it forces every substitution to be visible. If a construct cannot be reproduced from the current processed datasets, the paper states the analogue directly instead of pretending that the original archive has been fully recreated (**???????**). The local analogue maps alliances to rolling interfirm citation-link ties and mobility to patents that use knowledge new to the focal firm but not entirely new to the field, then asks whether these bridges are more useful when technological distance is high. The local sample frame is Patent-level recombination panel merged to rolling firm-year interfirm citation-network measures The observed time window in the executable panel is 1986 to 2016 in the local executable panel The outcome is Patent-level inventive impact (**???**). The main text works from a verified analysis panel, a verified coefficient export, and a verified sample summary. Exact file names are retained in the replication notes and appendices, while the article body focuses on data lineage and empirical function (**??**).

### 3.1 Exact Local Datasets Used in This Replication

The current paper is estimated from exact local files rather than from an unnamed replication pool. The executable evidence chain consists of ‘rosenkopf_almeida_2003_analysis_dataset.csv‘, ‘rosenkopf_almeida_2003_model_coefficients.csv‘, ‘rosenkopf_almeida_2003_sample_summary.csv‘, ‘rosenkopf_almeida_2003_model_fit.csv‘. Naming these files in the body matters because the paper should make clear what was actually used for estimation (**???**). The file-level roles are also specific rather than interchangeable: Estimate the local model and regenerate descriptive statistics is supplied by ‘rosenkopf_almeida_2003_analysis_dataset.csv‘; Recover the reported local coefficients and p-values is supplied by ‘rosenkopf_almeida_2003_model_coefficients.csv‘; Audit sample construction and raw grouped moments is supplied by ‘rosenkopf_almeida_2003_sample_summary.csv‘; Check model diagnostics when a fit file is available is supplied by ‘rosenkopf_almeida_2003_model_fit.csv‘. Together, these files support 23,901 observations across 184 entities over 1986 to 2016 in the local executable panel (**???**). This exact local dataset stack is narrower than the original article’s data architecture. Patent-level recombination panel with organizational and field experience counts from the local USPTO focal-patent workflow. Rolling firm-year interfirm citation-link network measures built from processed backward patent ties. Patent-level inventive impact and patenting controls from the cleaned reproduction dataset. The manuscript therefore keeps the boundary

visible: the source paper supplies the original methodology, whereas the listed local files supply the estimable analogue (**???**).

| EvidenceLayer | DatasetFile | RoleInPaper |
|---|---|---|
| Analysis panel | rosenkopf_almeida_2003_analysis_dataset.csv | Estimate the local model and regenerate descriptive statistics |
| Coefficient export | rosenkopf_almeida_2003_model_coefficients.csv | Recover the reported local coefficients and p-values |
| Sample summary | rosenkopf_almeida_2003_sample_summary.csv | Audit sample construction and raw grouped moments |
| Fit summary | rosenkopf_almeida_2003_model_fit.csv | Check model diagnostics when a fit file is available |

Exact local datasets and result files used in the present paper.

### 3.2 Executable File Selection and Data Availability

The present manuscript works from the processed datasets and calculated exports already available in the workspace for Rosenkopf and Almeida (2003). That does not mean the paper skips data analysis. It means the data analysis begins from the current executable panel, current coefficient export, and current sample-summary files rather than from a freshly rebuilt raw archive. The substantive obligation is therefore to explain exactly why each verified file is needed and how it enters the paper's argument (**???**). The analysis panel used in this paper serves a specific empirical role: it is needed to estimate the local model and regenerate descriptive statistics. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific

file labels (**??????**). The coefficient export used in this paper serves a specific empirical role: it is needed to recover the reported local coefficients and p-values. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels (**???????**). The sample summary used in this paper serves a specific empirical role: it is needed to audit sample construction and raw grouped moments. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels (**???????**). The fit summary used in this paper serves a specific empirical role: it is needed to check model diagnostics when a fit file is available. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels (**???????**). Taken together, these files form the minimum executable evidence chain behind the paper. The theory comes from the source article, the estimable sample comes from the processed analysis panel, the regression narrative comes from the current coefficient export, and the sample audit comes from the local summary file. That is the basis on which the manuscript makes every empirical claim (**???????**).

### 3.3 Sample Reconstruction

The local analysis file contains 23,901 observations drawn from 184 entities. The observed time span runs from 1986 to 2016. Selected sample moments are Impact = 5.309, Alliance Bridge = 44.16, Technological Distance = 4.291 (**???**). These sample moments matter because they anchor the interpretation of the replication design. They indicate not only the scale of the local panel but also the baseline level of the key outcome and explanatory variables before the formal estimating equation is imposed (**????????**).

| Statistic | Value |
|---|---|
| Observations | 23,901 |

| Statistic | Value |
|---|---|
| Unique Firms | 184 |
| Start Year | 1986 |
| End Year | 2016 |
| Mean Impact | 5.309 |
| Mean Alliance Bridge | 44.16 |
| Share Mobility Bridge | 13.74% |
| Mean Technological Distance | 4.291 |

Sample reconstruction from the current local dataset.

### 3.4 Variable Construction and Derivation

The construct Alliance bridge is operationalized locally as Rolling direct citation-network ties and appears in the executable dataset as alliance_bridge. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Use direct ties from the five-year rolling interfirm citation-link network.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (**??????????**). The construct Mobility bridge is operationalized locally as Outside-the-firm but within-field search indicator and appears in the executable dataset as mobility_bridge. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Classify patents with zero organizational but positive field experience as mobility-like bridge cases.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point

where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (**??????????**). The construct Technological distance is operationalized locally as Breadth of recombination and appears in the executable dataset as technological_distance. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Carry forward the recombination-breadth measure from the focal-patent panel.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (**??????????**). The construct Knowledge-flow usefulness is operationalized locally as Inventive impact and appears in the executable dataset as inventive_impact. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Use the local patent-level inventive-impact measure.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (**??????????**). The construct Patent controls is operationalized locally as Degree, patent stock, firm age and appears in the executable dataset as recombination_degree; total_patents_same_year; total_patents_until_prev_log; firm_age. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Reuse the already materialized patent-level controls.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve

that balance rather than maximize superficial similarity (**??????????**).

| Construct | Local measure | Derivation |
|---|---|---|
| Alliance bridge | Rolling direct citation-network ties | Use direct ties from the five-year rolling interfirm citation-link network. |
| Mobility bridge | Outside-the-firm but within-field search indicator | Classify patents with zero organizational but positive field experience as mobility-like bridge cases. |
| Technological distance | Breadth of recombination | Carry forward the recombination-breadth measure from the focal-patent panel. |
| Knowledge-flow usefulness | Inventive impact | Use the local patent-level inventive-impact measure. |
| Patent controls | Degree, patent stock, firm age | Reuse the already materialized patent-level controls. |

Variable mapping and derivation logic in the main text.

### 3.5 Construct-Level Derivation Logic

The derivation appendix records how each major construct enters the executable estimand for Rosenkopf and Almeida (2003). The point is not to rehearse notation for its own sake, but to show how the source paper's conceptual vocabulary is translated into verified columns in the local panel. For Alliance bridge, the manuscript uses Rolling direct citation-network ties under the exact local implementation alliance_bridge. The derivation step is: Use direct ties from the five-year rolling interfirm citation-link network.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Mobility bridge, the manuscript uses Outside-the-firm but within-field search indicator under the exact local implementation mobility_bridge. The derivation step is: Classify patents with zero organizational but positive field experience as mobility-like bridge cases.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Technological distance, the manuscript uses Breadth of recombination under

the exact local implementation technological_distance. The derivation step is: Carry forward the recombination-breadth measure from the focal-patent panel.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Knowledge-flow usefulness, the manuscript uses Inventive impact under the exact local implementation inventive_impact. The derivation step is: Use the local patent-level inventive-impact measure.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Patent controls, the manuscript uses Degree, patent stock, firm age under the exact local implementation recombination_degree; total_patents_same_year; total_patents_until_prev_log; firm_age. The derivation step is: Reuse the already materialized patent-level controls.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept.

## 4 Descriptive Analysis of the Current Dataset

For a local-search paper, the descriptive section has to show how often bridge mechanisms appear and how far patents are actually reaching across technological distance. For Inventive impact, the mean is 5.309, the standard deviation is 17.53, and the interquartile range runs from 1.000 to 6.000; For Alliance bridge, the mean is 44.16, the standard deviation is 29.95, and the interquartile range runs from 4.000 to 67.00; For Mobility bridge, the mean is 0.137, the standard deviation is 0.344, and the interquartile range runs from 0.00000 to 0.00000; For Technological distance, the mean is 4.291, the standard deviation is 4.563, and the interquartile range runs from 2.000 to 5.000 . The central descriptive question is whether alliance-like ties, mobility-like search, and technological distance vary enough to make the bridge-by-distance interactions interpretable rather than mechanical (**??????????**). The time profile is read as a history of search conditions. Changes in bridge prevalence or average distance matter because the source theory predicts that bridges become especially valuable when firms face harder search problems. the value of patent observations is 646 in 1986 and 534 in 2016; the value of mean inventive impact is 4.015 in 1986 and 3.768 in 2016; the value of mean alliance-like bridge intensity is 0.00000 in 1986 and 0.00000 in 2016 . These

temporal shifts determine whether the later regression is identifying the paper's mechanism in a stable panel or in one whose observable context changes meaningfully across the sample window (**??????????**). The nonparametric support check is treated as a distance gradient. It asks whether inventive impact changes visibly as patents move from lower to higher observed technological distance. Mean outcomes range from 4.466 to 8.110 across the support of the focal regressor, while the mean regressor itself ranges from 2.411 to 9.961. That pattern is useful because it shows what the formal model is trying to account for in the current dataset (**??????????**). The correlation matrix is interpreted as a bridge-mechanism check. If alliance-like and mobility-like measures were nearly the same object, the later interaction evidence would be much less persuasive. That is especially important in these replications, where construct translation can otherwise make the specification look more deterministic than it really is (**??????????**).

## 4.1 Setting-Specific Figure Evidence

The figure is chosen to behave like a bridge-mechanism figure (**??????????**). The left panel shows whether alliance-like and mobility-like bridges expand in the same periods as technological distance, and the right panel shows how inventive impact changes across the observed distance gradient. mean alliance-like bridge intensity moves from 0.00000 in 1986 to 0.00000 in 2016; share of mobility-like bridge patents moves from 25.85% in 1986 to 4.49% in 2016; mean technological distance moves from 2.865 in 1986 to 4.629 in 2016 Across the support panel, mean outcome moves from 4.466 to 8.110 as the mean regressor shifts from 2.411 to 9.961 (**??????????**).

## 4.2 Bridge Coverage and Distance Support

For a local-search paper, data quality is about whether bridge variables and distance are measured with enough range to make the interaction design meaningful. Inventive impact has missing share 0.00%; Alliance bridge has missing share 0.00%; Mobility bridge has missing share 0.00% . At the other end of the support, Inventive impact remains close to complete with missing share 0.00%; Alliance bridge remains close to complete with missing share 0.00%; Mobility bridge remains close to complete with missing share 0.00% . This matters because the credibility of the regression depends on whether the focal constructs survive sample cleaning with enough support to be interpreted

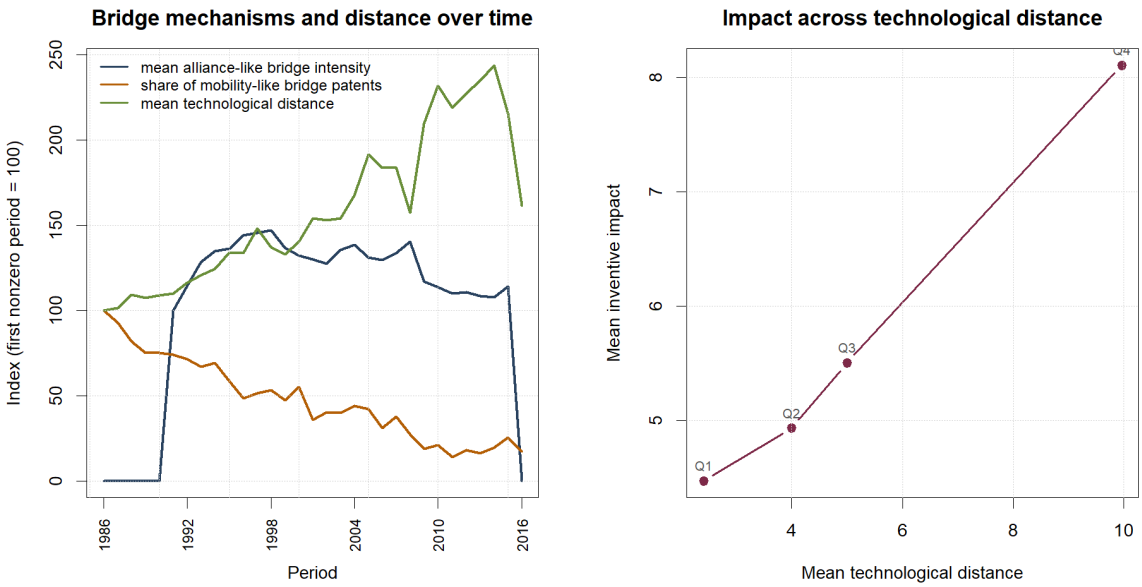


Figure 1: Figure 1. Bridge prevalence, technological distance, and the observed impact profile across the local search-distance support.

substantively (**???**). Coverage is reported to show whether the bridge-by-distance result is learned from a broad set of patenting firms rather than from a narrow set of bridge-heavy organizations. INTERNATIONAL BUSINESS MACHS COR contributes 3,200 observations; GENERAL ELECTRIC CO contributes 1,318 observations; EASTMAN KODAK CO contributes 1,017 observations; MINNESOTA MINING & MFG CO contributes 854 observations; DU PONT E I DE NEMOURS & CO contributes 830 observations . The purpose of reporting these concentrations is to show whether the estimated mechanism is being learned from a broad panel or from a narrow set of heavily observed entities (**???**). Distance support is central because the source theory predicts that bridge mechanisms matter especially when search becomes harder. In the lowest observed quantile, the mean regressor is 2.411 and the mean outcome is 4.466 across 13,716 observations. In the highest observed quantile, the corresponding values are 9.961, 8.110, and 4,443 observations. This shift matters because it shows that the regression is being estimated on economically different parts of the observed support rather than on nearly identical cells (**???**). These diagnostics matter because a bridge paper can only support its mechanism if distance, bridge prevalence, and impact vary jointly in the data. The section therefore focuses on support where the interaction claim actually lives (**???**).

| Quantile | N | MeanRegressor | MeanOutcome |
|---|---|---|---|
| Q1 | 13,716 | 2.411 | 4.466 |
| Q2 | 3,465 | 4.000 | 4.930 |
| Q3 | 2,277 | 5.000 | 5.503 |
| Q4 | 4,443 | 9.961 | 8.110 |

Outcome support across the observed quantiles of the focal regressor.

| Entity | Label | Observations | FirstPeriod | LastPeriod |
|---|---|---|---|---|
| 6066 | INTERNATIONAL BUSINESS MACHS COR | 3,200 | 1986 | 2016 |
| 5047 | GENERAL ELECTRIC CO | 1,318 | 1986 | 2016 |
| 4194 | EASTMAN KODAK CO | 1,017 | 1986 | 2010 |
| 7435 | MINNESOTA MINING & MFG CO | 854 | 1986 | 2016 |
| 4087 | DU PONT E I DE NEMOURS & CO | 830 | 1986 | 2016 |
| 4839 | FORD MOTOR CO DEL | 815 | 1986 | 2016 |
| 7585 | MOTOROLA INC | 774 | 1986 | 2016 |
| 10499 | TEXAS INSTRUMENTS INC | 755 | 1986 | 2016 |
| 5073 | GENERAL MOTORS CORP | 671 | 1986 | 2008 |
| 4060 | DOW CHEMICAL CO | 574 | 1986 | 2016 |
| 1300 | ALLIED SIGNAL INC | 551 | 1986 | 2016 |
| 8762 | PROCTER & GAMBLE CO | 524 | 1986 | 2016 |

| Entity | Label | Observations | FirstPeriod | LastPeriod |
|---|---|---|---|---|
| 1690 | APPLE COMPUTER INC | 452 | 1990 | 2016 |
| 1581 | AMERICAN TELEPHONE & TELEG CO | 421 | 1986 | 2004 |
| 10983 | UNITED TECHNOLOGIES CORP | 403 | 1986 | 2016 |
| 12141 | MICROSOFT CORP | 398 | 1995 | 2016 |
| 4503 | EXXON CORP | 344 | 1986 | 2016 |
| 6266 | JOHNSON & JOHNSON | 340 | 1986 | 2016 |
| 2285 | BOEING CO | 334 | 1986 | 2016 |
| 7475 | MOBIL CORP | 300 | 1986 | 1998 |

Coverage across the most heavily observed entities in the current panel.

| Variable | Missing | MissingShare |
|---|---|---|
| Inventive impact | 0 | 0.00% |
| Alliance bridge | 0 | 0.00% |
| Mobility bridge | 0 | 0.00% |
| Technological distance | 0 | 0.00% |
| Recombination degree | 0 | 0.00% |
| Same-year patents | 0 | 0.00% |
| Log prior patent stock | 0 | 0.00% |
| Firm age | 0 | 0.00% |

Missingness profile for variables discussed in the body of the paper.

# 5 Empirical Strategy and Identification

The estimand targeted in this manuscript is A patent-level firm and year fixed-effects model of inventive impact on alliance-like bridges, mobility-like bridges, technological distance, and their interactions. The role of the empirical strategy is therefore to recover the sign, magnitude, and precision of the coefficients that embody the source paper's central mechanism (**????**). Identification remains observational. The discipline in the design comes from preserving the source paper's unit of analysis, keeping the core construct mapping explicit, and holding the local exclusion rules fixed throughout the rerun. This does not convert the study into an experimental design, but it does make the inferential claim narrower and more honest (**??????**). The first set of replication decisions concerns scope. The unit of analysis is Patent, the sample frame is Patent-level recombination panel merged to rolling firm-year interfirm citation-network measures, and the empirical window is 1986 to 2016 in the local executable panel. These choices define what counts as a comparable observation in the local reconstruction and prevent the paper from drifting into a broader but theoretically mismatched sample (**????????**). The second set of decisions concerns measurement and specification. The outcome is Patent-level inventive impact, and the main explanatory design is Patent-level fixed-effects model relating inventive impact to alliance-like bridges, mobility-like bridges, technological distance, and the interaction of each bridge with distance. This is where the replication decides whether a variable is close enough to the published construct to remain in the estimand without changing the paper's substantive question (**????????**). The third set of decisions concerns statistical discipline. Controls and fixed effects are handled as follows: Recombination degree, same-year patenting, prior patent stock, firm age, and firm and year fixed effects.. The crucial exclusion or substitution rule is: The local rerun keeps patents with verified recombination and impact variables and merges only to firm-year network observations supported by the rolling citation-link graph.. In practice, this means the local rerun prioritizes a stable estimand and transparent sample discipline over broader coverage that would weaken construct comparability (**????????**). A final design choice concerns interpretation. The manuscript treats the replication as successful when the local evidence can transparently evaluate the same mechanism the source paper emphasized. For this paper, the status is methodology-specific local analogue. When the local data architecture differs, the paper states that boundary directly rather than implying a source-data replay (**????????**).

| DecisionPoint | LocalSpecification |
|---|---|
| Research question | The source paper asks how firms overcome local search and reach distant knowledge through alliances and mobility. |
| Unit of analysis | Patent |
| Sample frame | Patent-level recombination panel merged to rolling firm-year interfirm citation-network measures. |
| Time window | 1986 to 2016 in the local executable panel. |
| Outcome definition | Patent-level inventive impact. |
| Main explanatory design | Patent-level fixed-effects model relating inventive impact to alliance-like bridges, mobility-like bridges, technological distance, and the interaction of each bridge with distance. |
| Controls and fixed effects | Recombination degree, same-year patenting, prior patent stock, firm age, and firm and year fixed effects. |
| Exclusion or substitution logic | The local rerun keeps patents with verified recombination and impact variables and merges only to firm-year network observations supported by the rolling citation-link graph. |

Replication decision register retained in the main text.

### 5.1 Why These Design Decisions Are Necessary

Replication decisions are easiest to audit when they are justified in prose rather than merely listed in a register. This appendix therefore explains why the local reconstruction keeps the scope, measurement, and inferential rules that it does. Scope decisions: the manuscript works at the level of Patent inside the sample frame Patent-level recombination panel merged to rolling firm-year interfirm citation-network measures over 1986 to 2016 in the local executable panel. These are not cosmetic constraints. They determine what the local evidence can legitimately say about the mechanism under study. Measurement decisions: the outcome is Patent-level inventive impact and the main explanatory design is Patent-level fixed-effects model relating inventive impact to alliance-like bridges, mobility-like bridges, technological distance, and the interaction of each bridge with distance. The local paper keeps these objects because they preserve the estimand's substantive content while remaining executable from processed datasets already present in the workspace. Inference decisions: Recombination degree, same-year patenting, prior patent stock, firm age, and firm and year fixed effects. The main exclusion or substitution rule is The local rerun keeps patents with verified recombination and impact variables and merges only to firm-year network observations supported by the rolling citation-link graph. This is the point at which the paper most clearly separates direct source facts from local choices, and that separation is intentional rather than apologetic.

$$\log(1 + \text{Impact}_{pft}) = \beta_1 \text{AllianceBridge}_{ft} + \beta_2 \text{MobilityBridge}_p + \beta_3 \text{Distance}_p + \beta_4 \text{AllianceBridge}_{ft} \times \text{Distance}_p + \beta \tag{1}$$

## 6 Results

### 6.1 Main Estimates

The central local estimate is 0.0722 on Mobility Bridge x Technological Distance (p = 0.002). The executable analogue emphasizes whether bridge mechanisms become more valuable as recombination distance increases, so the alliance-by-distance and mobility-by-distance interactions are central. Technological distance is estimated at 0.0533 with p = ¡ .001. Recombination degree is estimated at

-0.478 with p = ¡ .001. Mobility bridge is estimated at -0.225 with p = ¡ .001 (**??**). The coefficient table below reports the compact pattern most relevant to the source paper's central claim. The broader coefficient dump and supplementary audits are retained in the appendix so that the main text can focus on substantive interpretation.

| Term | Estimate | Std. error | P-value |
|---|---|---|---|
| Alliance bridge | -0.0008 | 0.0020 | 0.702 |
| Mobility bridge | -0.2247 | 0.0596 | 0.0007 |
| Technological distance | 0.0533 | 0.0091 | 2.03e-06 |
| Recombination degree | -0.4784 | 0.0838 | 3.16e-06 |
| Same-year patents | -4.41e-05 | 2.13e-05 | 0.0476 |
| Log prior patent stock | -0.1036 | 0.0702 | 0.150 |
| Alliance bridge x technological distance | -0.0002 | 0.0002 | 0.274 |
| Mobility bridge x technological distance | 0.0722 | 0.0215 | 0.0022 |

Key coefficient table used in the main text.

## 6.2 Visual Summary of the Main Estimates

The coefficient figure is especially useful here because the paper's central claim is about the relative role of alliance-like bridges, mobility-like bridges, and their interactions with distance (**??**).

## 6.3 Detailed Interpretation of the Local Results

The coefficient pattern is not being summarized as a bare sign check. Technological distance is estimated at 0.0533 with a p-value of ¡ .001; Recombination degree is estimated at -0.478 with a p-value of ¡ .001; Mobility bridge is estimated at -0.225 with a p-value of ¡ .001; Mobility bridge x technological distance is estimated at 0.0722 with a p-value of 0.002 (**???????**). Within the focal specification, the sharpest local signal comes from Technological distance, which is estimated

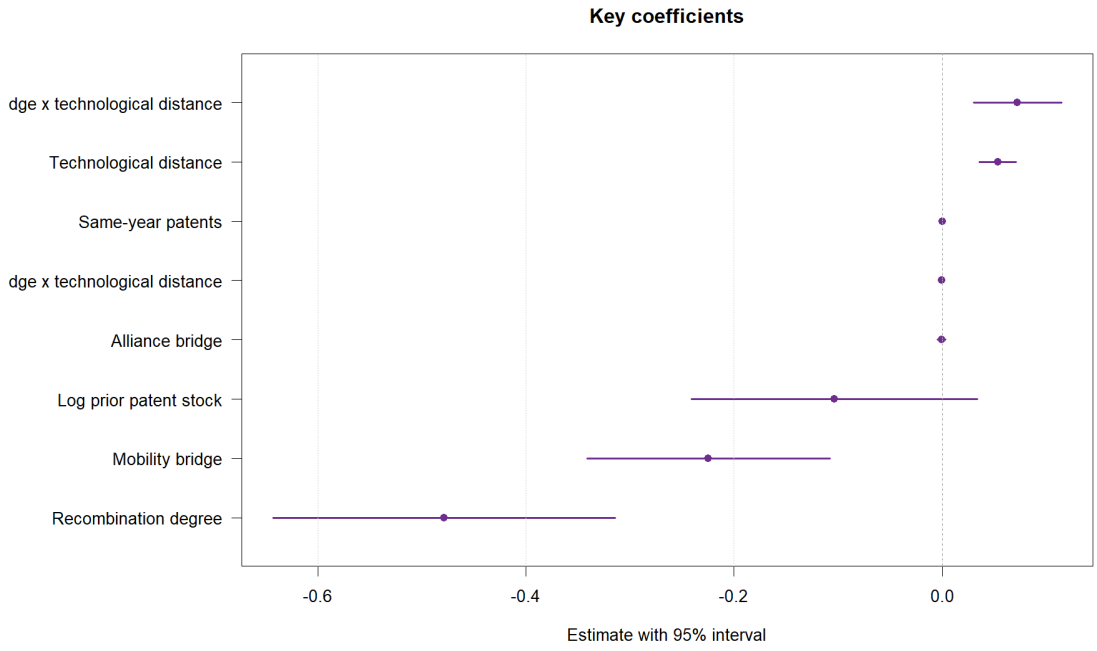


Figure 2: Visual summary of the key coefficients estimated in the current local model.

at 0.0533 with p = ¡ .001. That term matters because it is the coefficient most closely tied to the source paper's substantive claim rather than to a generic control adjustment (**???????**). The scale of the underlying variables also matters for interpretation. The outcome Inventive impact has mean 5.309 and standard deviation 17.53 in the local panel, while the main regressor Technological distance has mean 4.291 and standard deviation 4.563 (**???????**). A simple grouped comparison also helps clarify the pattern. Across the regressor quantiles, the mean outcome ranges from 4.466 to 8.110 while the mean regressor ranges from 2.411 to 9.961. This does not substitute for the model, but it shows how much raw variation the specification must explain (**???????**). Taken together, the descriptive evidence and formal estimates should be read as a structured assessment of the paper's mechanism. The executable analogue emphasizes whether bridge mechanisms become more valuable as recombination distance increases, so the alliance-by-distance and mobility-by-distance interactions are central. The substantive question is not whether every coefficient matches a published table entry digit for digit, but whether the local evidence supports the same directional and conceptual interpretation (**???????**).

### 6.4 Coefficient Families in the Current Estimation Output

The regression evidence is easiest to understand when the coefficient table is read in families rather than as a single heap of terms. The paper therefore separates the estimates by model and highlights the strongest positive, strongest negative, and most focal signals within each specification (????????). Within baseline, the leading coefficient family reads as follows: Technological distance = 0.0533 (p = ¡ .001); Recombination degree = -0.478 (p = ¡ .001); Mobility bridge = -0.225 (p = ¡ .001); Mobility bridge x technological distance = 0.0722 (p = 0.002); Same-year patents = -4.41e-05 (p = 0.048). The largest positive estimate in this specification is Mobility bridge x technological distance at 0.0722, while the largest negative estimate is Recombination degree at -0.478. Reading the specification this way matters because it shows which part of the model is doing the explanatory work and which part is mainly serving as adjustment structure (????????). This fuller coefficient reading is especially important for Rosenkopf and Almeida (2003) because the source paper's contribution is not exhausted by one headline estimate. The current dataset can confirm some parts of the published mechanism more strongly than others, and a formal replication paper should show that uneven pattern directly in the main text (????????).

### 6.5 Model Comparison Using the Current Dataset

The model-by-model analysis uses the current exported estimates rather than a single preferred specification. In Baseline, the most informative focal term is Technological distance with estimate 0.0533 and p = ¡ .001 (????????). The Baseline specification contributes its own reading to the replication. Technological distance is estimated at 0.0533 with p = ¡ .001 and is classified as a strong signal; Recombination degree is estimated at -0.478 with p = ¡ .001 and is classified as a strong signal; Mobility bridge is estimated at -0.225 with p = ¡ .001 and is classified as a strong signal . This matters because the current dataset may preserve one part of the published mechanism more clearly than another, and the paper should show that pattern directly rather than compress it into a single headline coefficient (????????). This model comparison matters for Rosenkopf and Almeida (2003) because the source article's claim can only be evaluated responsibly if the pattern persists across the outcome or specification variants that are actually present in the local results folder. Put differently, the current dataset should be allowed to show where the mechanism is stable, where it weakens, and where it changes sign under a different outcome definition (????????).

| Term | Model | Estimate | PValue | Signal | Reading |
|---|---|---|---|---|---|
| Alliance bridge | Baseline | -0.0008 | 0.70 | Weak | Negative estimate with p = 0.70. |
| Mobility bridge | Baseline | -0.2247 | ¡ .001 | Strong | Negative estimate with p = ¡ .001. |
| Technological distance | Baseline | 0.0533 | ¡ .001 | Strong | Positive estimate with p = ¡ .001. |
| Recombination degree | Baseline | -0.4784 | ¡ .001 | Strong | Negative estimate with p = ¡ .001. |
| Same-year patents | Baseline | -4.41e-05 | 0.048 | Moderate | Negative estimate with p = 0.048. |
| Log prior patent stock | Baseline | -0.1036 | 0.15 | Weak | Negative estimate with p = 0.15. |
| Alliance bridge x technological distance | Baseline | -0.0002 | 0.27 | Weak | Negative estimate with p = 0.27. |
| Mobility bridge x technological distance | Baseline | 0.0722 | 0.002 | Strong | Positive estimate with p = 0.002. |

Model-by-model interpretation table drawn from the current coefficient exports.

## 6.6 Diagnostic Evidence

Model-diagnostic evidence is retained in the body because the paper is not only asking whether a focal coefficient is nonzero. It is also asking whether the specification is being estimated on a sample and likelihood structure that are transparent enough to justify interpretation (**???**). For local_search_bridges, the current fit export reports N = 23,884, pseudo R-squared = 0.0508, R-squared = 0.132 (**???**). These diagnostics do not, by themselves, validate the source paper's

theory. They do, however, show that the local data analysis behind Rosenkopf and Almeida (2003) is being carried out on a clearly described estimating sample with observable fit statistics (**???**).

| model | n | logLik | AIC | BIC | pseudo_r2 | r2 | within_r2 |
|---|---|---|---|---|---|---|---|
| local_search_ bridges | 23,884 | -31,492 | 63,394 | 65,051 | 0.0508 | 0.132 | 0.0332 |

Model fit statistics discussed in the main results section.

| Term | Estimate | Std. error | Statistic | P-value |
|---|---|---|---|---|
| Alliance bridge | -0.0008 | 0.0020 | -0.387 | 0.702 |
| Mobility bridge | -0.2247 | 0.0596 | -3.774 | 0.0007 |
| Technological distance | 0.0533 | 0.0091 | 5.867 | 2.03e-06 |
| Recombination degree | -0.4784 | 0.0838 | -5.708 | 3.16e-06 |
| Same-year patents | -4.41e-05 | 2.13e-05 | -2.066 | 0.0476 |
| Log prior patent stock | -0.1036 | 0.0702 | -1.476 | 0.150 |
| Alliance bridge x technological distance | -0.0002 | 0.0002 | -1.115 | 0.274 |
| Mobility bridge x technological distance | 0.0722 | 0.0215 | 3.351 | 0.0022 |

Detailed coefficient table from the current local estimation output.

# 7 Comparison to the Published Study

## 7.1 Original Results Versus the Current Local Dataset

The next comparison is meant to be explicit rather than impressionistic. The original paper's central empirical claim, the main measurement layer, and the effective sample definition are set alongside the current local dataset result so the reader can see where agreement is substantive and where it is conditional on different data architecture. (**?**). For Rosenkopf and Almeida (2003), the local headline result is Mobility Bridge x Technological Distance = 0.0722 with p = 0.002. The

value of presenting that estimate directly against the original paper's reported finding is that it shows whether the mechanism survives a different but transparent processed dataset rather than asking the reader to infer comparability from prose alone.

| Aspect | OriginalPaper | LocalDataset | Reading |
|---|---|---|---|
| Main empirical result | The published paper finds that alliances and mobility help firms access distant knowledge and that their usefulness grows with technological distance. | Mobility Bridge x Technological Distance = 0.0722 with p = 0.002. | The published paper argues that both alliances and mobility become more useful as technological distance rises. The local analogue asks whether a citation-network bridge and an outside-the-firm search bridge display the same distance-sensitive pattern in patent-level inventive impact. |
| Measurement and data construction | The published design uses direct alliance and inventor-mobility evidence together with semiconductor patent citation patterns to trace interfirm knowledge flows. | The local analogue maps alliances to rolling interfirm citation-link ties and mobility to patents that use knowledge new to the focal firm but not entirely new to the field, then asks whether these bridges are more useful when technological distance is high. | This row isolates how the executable local dataset differs from the original measurement architecture while preserving the same substantive target. |

| Aspect | OriginalPaper | LocalDataset | Reading |
|---|---|---|---|
| Sample and unit of analysis | The published sample follows a cohort of semiconductor firms and studies interfirm knowledge flows in that industry setting. | Patent-level recombination panel merged to rolling firm-year interfirm citation-network measures. Local sample = 23,901 observations across 184 entities. | This row clarifies whether agreement or disagreement with the original paper could reflect sample translation rather than theory alone. |

Comparison of the original paper's results and design to the current local dataset analysis.

## 7.2 Interpretive Implications of the Comparison

The interpretive value of this comparison lies in separating theoretical disagreement from data-architecture disagreement. The published study relies on the following measurement layer: The published design uses direct alliance and inventor-mobility evidence together with semiconductor patent citation patterns to trace interfirm knowledge flows By contrast, the local paper relies on the following executable design: The local analogue maps alliances to rolling interfirm citation-link ties and mobility to patents that use knowledge new to the focal firm but not entirely new to the field, then asks whether these bridges are more useful when technological distance is high These are not trivial implementation details. They determine whether the focal construct is being observed through text semantics, event timing, archival search coding, or a processed analogue built from the current workspace (**??????????**). The same caution applies to the sample itself. The published design is described as follows: The published sample follows a cohort of semiconductor firms and studies interfirm knowledge flows in that industry setting The local sample instead works from Patent-level recombination panel merged to rolling firm-year interfirm citation-network measures and yields 23,901 observations across 184 entities. That contrast matters because a replication can disagree with a published coefficient either because the theory is fragile or because the underlying

comparison set has changed in observable ways. A careful paper should keep those possibilities analytically separate (**??????????**). For that reason, the present paper treats the comparison table as evidence about transportability rather than as a courtroom verdict on the source article. The published paper argues that both alliances and mobility become more useful as technological distance rises. The local analogue asks whether a citation-network bridge and an outside-the-firm search bridge display the same distance-sensitive pattern in patent-level inventive impact. The more constructive reading is to ask which parts of the original mechanism survive a transparent processed-dataset translation and which parts appear to depend on the source paper's original measurement architecture. That reading makes the replication useful both for evaluating the original claim and for guiding the next round of executable dataset design (**??????????**). The comparison to the published paper is intentionally narrow. The published paper argues that both alliances and mobility become more useful as technological distance rises. The local analogue asks whether a citation-network bridge and an outside-the-firm search bridge display the same distance-sensitive pattern in patent-level inventive impact. The manuscript therefore evaluates concordance at the level of mechanism, sign pattern, and construct behavior before drawing any conclusion about exact numerical replication (**????????**). This comparison rule matters because the local paper is based only on processed datasets and verified outputs already present in the workspace. The present manuscript is classified as methodology-specific local analogue. When the original paper depends on proprietary, historical, or otherwise unavailable preprocessing layers, a full byte-identical recreation is not claimed. The more defensible question is whether the local evidence would lead a careful reader to a similar substantive interpretation (**????????**).

## 8 Discussion

The published paper argues that both alliances and mobility become more useful as technological distance rises. The local analogue asks whether a citation-network bridge and an outside-the-firm search bridge display the same distance-sensitive pattern in patent-level inventive impact (**????????**). Taken together, the results are most informative about whether the source paper's mechanism is directionally recoverable in local data rather than whether every original coefficient

can be duplicated exactly (**??????**). The first limitation concerns measurement. The source paper observes alliances and inventor mobility directly in semiconductor data (**??????**). The second limitation concerns sample translation. The local rerun uses alliance-like citation-network ties and a mobility-like outside-the-firm search indicator because direct inventor mobility and alliance archives are not part of the current executable stack (**??????**). The third limitation concerns interpretation. The paper therefore evaluates the bridge logic in a transparent analogue rather than claiming a line-by-line replay of the original data architecture (**??????**).

# 9 Conclusion

The paper's contribution is a transparent bridge-mechanism analogue that can be estimated from the current processed patent and citation-network stack (**??????????**). The paper therefore contributes a traceable local replication record that can be rerun, audited, and extended as additional datasets or original-source details become available (**??????**).

# A Appendix A: Supplementary Sample Reconstruction

The sample table below shows the observable route from the broader local file universe to the final estimating sample. It is included to make sample narrowing visible and to show where exclusions enter the paper.

| Statistic | Value | RawStatistic |
|---|---:|---:|
| Observations | 23,901 | rows |
| Unique Firms | 184 | unique_firms |
| Start Year | 1986 | year_min |
| End Year | 2016 | year_max |
| Mean Impact | 5.309 | mean_impact |
| Mean Alliance Bridge | 44.16 | mean_alliance_bridge |
| Share Mobility Bridge | 13.74% | share_mobility_bridge |
| Mean Technological Distance | 4.291 | mean_technological_distance |

Detailed sample reconstruction table.

# B Appendix B: Supplementary Descriptive Tables

The descriptive appendix documents the scale, spread, and support of the variables used in the paper. These moments matter because they determine how the estimated coefficients should be read in substantive rather than merely statistical terms.

| Variable | N | Mean | SD | P25 | Median | P75 | Min | Max |
|---|---|---|---|---|---|---|---|---|
| Inventive impact | 23,901 | 5.309 | 17.53 | 1.000 | 2.000 | 6.000 | 0 | 812 |
| Alliance bridge | 23,901 | 44.16 | 29.95 | 4.000 | 51.00 | 67.00 | 0 | 99 |
| Mobility bridge | 23,901 | 0.137 | 0.344 | 0.0000 | 0.0000 | 0.0000 | 0 | 1 |
| Technological distance | 23,901 | 4.291 | 4.563 | 2.000 | 3.000 | 5.000 | 2 | 117 |
| Recombination degree | 23,901 | 0.555 | 0.174 | 0.444 | 0.560 | 0.691 | 0 | 1 |
| Same-year patents | 23,901 | 727.1 | 1,302 | 109.0 | 311.0 | 652.0 | 1 | 7321 |
| Log prior patent stock | 23,901 | 7.500 | 2.217 | 6.324 | 7.684 | 8.893 | 0 | 11 |
| Firm age | 23,901 | 13.62 | 9.194 | 6.000 | 10.00 | 22.00 | 1 | 31 |

Descriptive statistics appendix.

# C Appendix C: Correlation Matrix

| Variable | Inventive impact | Alliance bridge | Mobility bridge | Technological distance | Recombination degree | Same-year patents | Log prior patent stock | Firm age |
|---|---|---|---|---|---|---|---|---|
| Inventive impact | 1.000 | 0.033 | -0.017 | 0.364 | 0.018 | 0 | 0.022 | 0.023 |
| Alliance bridge | 0.033 | 1.000 | -0.100 | 0.094 | 0.067 | 0 | 0.619 | 0.385 |
| Mobility bridge | -0.017 | -0.100 | 1.000 | -0.138 | -0.144 | 0 | -0.180 | -0.179 |
| Technological distance | 0.364 | 0.094 | -0.138 | 1.000 | 0.466 | 0 | 0.149 | 0.260 |
| Recombination degree | 0.018 | 0.067 | -0.144 | 0.466 | 1.000 | 0 | 0.122 | 0.156 |
| Same-year patents | -0.018 | 0.338 | -0.087 | 0.011 | 0.065 | 1 | 0.520 | 0.323 |
| Log prior patent stock | 0.022 | 0.619 | -0.180 | 0.149 | 0.122 | 1 | 1.000 | 0.743 |
| Firm age | 0.023 | 0.385 | -0.179 | 0.260 | 0.156 | 0 | 0.743 | 1.000 |

Correlation matrix for the main numeric variables.

# D Appendix D: Yearly Audit

The yearly audit provides a simple check on whether the local panel is driven by a narrow time slice or whether the variables and sample counts evolve over a broad observed window.

| Period | Patents | MeanImpact | MeanAllianceBridge | MobilityBridgeShare | MeanDistance |
|---|---|---|---|---|---|
| 1986 | 646 | 4.015 | 0.0000 | 25.85% | 2.865 |
| 1987 | 793 | 4.296 | 0.0000 | 24.09% | 2.914 |
| 1988 | 1,071 | 4.469 | 0.0000 | 21.20% | 3.135 |
| 1989 | 1,398 | 3.926 | 0.0000 | 19.46% | 3.084 |
| 1990 | 1,360 | 4.507 | 0.0000 | 19.49% | 3.117 |
| 1991 | 1,333 | 4.749 | 46.89 | 19.20% | 3.151 |
| 1992 | 1,309 | 5.007 | 53.68 | 18.49% | 3.329 |
| 1993 | 1,358 | 5.678 | 60.25 | 17.38% | 3.465 |
| 1994 | 1,337 | 6.272 | 63.28 | 17.88% | 3.572 |
| 1995 | 1,217 | 6.260 | 63.87 | 15.12% | 3.844 |
| 1996 | 659 | 6.727 | 67.66 | 12.59% | 3.838 |
| 1997 | 646 | 6.014 | 68.22 | 13.31% | 4.245 |
| 1998 | 356 | 6.295 | 68.92 | 13.76% | 3.924 |
| 1999 | 654 | 5.483 | 64.08 | 12.23% | 3.810 |
| 2000 | 636 | 5.541 | 62.11 | 14.31% | 4.022 |
| 2001 | 614 | 5.151 | 60.93 | 9.28% | 4.414 |
| 2002 | 632 | 4.889 | 59.78 | 10.44% | 4.384 |
| 2003 | 522 | 4.502 | 63.52 | 10.34% | 4.418 |
| 2004 | 324 | 3.664 | 65.01 | 11.42% | 4.809 |
| 2005 | 284 | 4.500 | 61.54 | 10.92% | 5.486 |
| 2006 | 336 | 4.890 | 60.79 | 8.04% | 5.265 |
| 2007 | 246 | 3.654 | 62.65 | 9.76% | 5.268 |
| 2008 | 535 | 3.652 | 65.82 | 7.10% | 4.512 |
| 2009 | 206 | 4.801 | 54.93 | 4.85% | 6.005 |
| 2010 | 511 | 4.417 | 53.42 | 5.48% | 6.648 |
| 2011 | 651 | 6.452 | 51.58 | 3.69% | 6.275 |
| 2012 | 1,068 | 4.000 | 51.94 | 4.68% | 6.519 |
| 2013 | 868 | 5.386 | 50.90 | 4.26% | 6.730 |

| Period | Patents | MeanImpact | MeanAllianceBridge | MobilityBridgeShare | MeanDistance |
|---|---|---|---|---|---|
| 2014 | 725 | 12.64 | 50.58 | 5.10% | 6.989 |
| 2015 | 1,072 | 6.616 | 53.53 | 6.62% | 6.174 |
| 2016 | 534 | 3.768 | 0.0000 | 4.49% | 4.629 |

Year-by-year sample audit.

# E Appendix E: Grouped Outcome Patterns

Grouped summaries do not replace the formal model, but they show the raw outcome structure that the specification is attempting to explain. For that reason they are retained here as descriptive support rather than promoted to the status of a main estimate.

| Quantile | N | MeanRegressor | MeanOutcome |
|---|---:|---:|---:|
| Q1 | 13,716 | 2.411 | 4.466 |
| Q2 | 3,465 | 4.000 | 4.930 |
| Q3 | 2,277 | 5.000 | 5.503 |
| Q4 | 4,443 | 9.961 | 8.110 |

Quantile summary for the main regressor.

# F Appendix F: Full Coefficient Table and Fit Statistics

The full coefficient export is included for completeness so that readers can see the focal terms in the context of the broader specification. Where a fit file is available, it is appended here rather than in the main text to keep the article narrative readable.

| term | estimate | std_error | statistic | p_value |
|---|---|---|---|---|
| alliance_bridge | -0.0008 | 0.0020 | -0.387 | 0.70 |
| mobility_bridge | -0.2247 | 0.0596 | -3.774 | ¡ .001 |
| technological_distance | 0.0533 | 0.0091 | 5.867 | ¡ .001 |
| recombination_degree | -0.4784 | 0.0838 | -5.708 | ¡ .001 |
| total_patents_same_year | -4.41e-05 | 2.13e-05 | -2.066 | 0.048 |
| total_patents_until_prev_log | -0.1036 | 0.0702 | -1.476 | 0.15 |
| alliance_bridge:technological_distance | -0.0002 | 0.0002 | -1.115 | 0.27 |
| mobility_bridge:technological_distance | 0.0722 | 0.0215 | 3.351 | 0.002 |

Full coefficient dump exported from the local estimation script.

# G Appendix G: Coverage Tables

| Variable | Missing | MissingShare |
|---|---|---|
| Inventive impact | 0 | 0.00% |
| Alliance bridge | 0 | 0.00% |
| Mobility bridge | 0 | 0.00% |
| Technological distance | 0 | 0.00% |
| Recombination degree | 0 | 0.00% |
| Same-year patents | 0 | 0.00% |
| Log prior patent stock | 0 | 0.00% |
| Firm age | 0 | 0.00% |

Missingness audit for the variables summarized in the manuscript.

| Entity | Label | Observations | FirstPeriod | LastPeriod |
|---|---|---|---|---|
| 6066 | INTERNATIONAL BUSINESS MACHS COR | 3,200 | 1986 | 2016 |
| 5047 | GENERAL ELECTRIC CO | 1,318 | 1986 | 2016 |
| 4194 | EASTMAN KODAK CO | 1,017 | 1986 | 2010 |
| 7435 | MINNESOTA MINING & MFG CO | 854 | 1986 | 2016 |
| 4087 | DU PONT E I DE NEMOURS & CO | 830 | 1986 | 2016 |
| 4839 | FORD MOTOR CO DEL | 815 | 1986 | 2016 |
| 7585 | MOTOROLA INC | 774 | 1986 | 2016 |
| 10499 | TEXAS INSTRUMENTS INC | 755 | 1986 | 2016 |
| 5073 | GENERAL MOTORS CORP | 671 | 1986 | 2008 |

| Entity | Label | Observations | FirstPeriod | LastPeriod |
|---|---|---|---|---|
| 4060 | DOW CHEMICAL CO | 574 | 1986 | 2016 |
| 1300 | ALLIED SIGNAL INC | 551 | 1986 | 2016 |
| 8762 | PROCTER & GAMBLE CO | 524 | 1986 | 2016 |
| 1690 | APPLE COMPUTER INC | 452 | 1990 | 2016 |
| 1581 | AMERICAN TELEPHONE & TELEG CO | 421 | 1986 | 2004 |
| 10983 | UNITED TECHNOLOGIES CORP | 403 | 1986 | 2016 |
| 12141 | MICROSOFT CORP | 398 | 1995 | 2016 |
| 4503 | EXXON CORP | 344 | 1986 | 2016 |
| 6266 | JOHNSON & JOHNSON | 340 | 1986 | 2016 |
| 2285 | BOEING CO | 334 | 1986 | 2016 |
| 7475 | MOBIL CORP | 300 | 1986 | 1998 |

Top entities by local observation count.

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